Project Design Phase
Solution Architecture

Date	12 October 2026
Team ID	SWUID2026403463
Project Name	HotelPro - Online Hotel Room Booking & Reservation Management System
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for **HotelPro** ensures a robust, user-friendly and scalable platform that connects guests with hotels and their management teams. It follows a layered three-tier design: a React single-page application, a stateless Spring Boot REST API secured with JWT and role-based access, and a relational database, supported by Redis for room holds and caching. The architecture focuses on responsive design, conflict-free room booking, secure mock payments and timely notifications to ensure a smooth booking journey.

- Seamless hotel discovery, room selection and online booking
- Real-time room availability with atomic room holds to prevent double booking
- Scalable user authentication and role-based authorization (guest, hotel manager, front-desk staff, admin)
- Secure and trackable payments, cancellations and refunds through a mock gateway
- Reliable QR booking vouchers, in-app and email notifications, and admin analytics

Example - Solution Architecture Diagram:

